

Analysis group meeting

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10/22/24



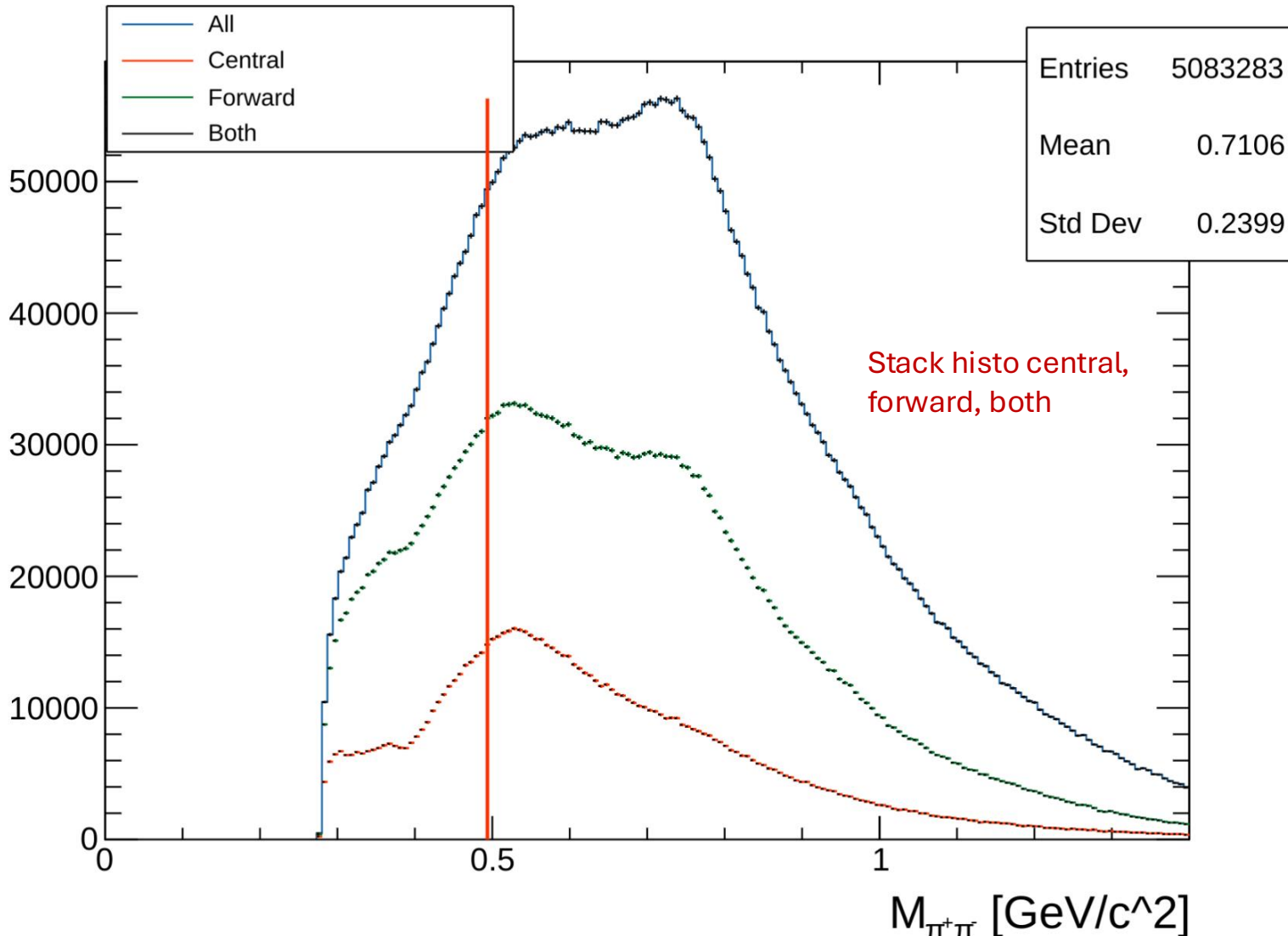
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Cooking chef

- Pass0v5 is done, timelines are availables.
 - rf calibration is done.
-
- Skim data are produced I need to copy them from volatile to scratch

RG-D Analysis: K^0

- Find K^0 but only in the central detectors.



Using pass0v4 LD2 data, all runs

Electron :

$-5 < \chi^2_{\text{pid}} < 5$

$-15 < v_z < 5 \text{ cm}$

Event:

$Z_h < 0.7$

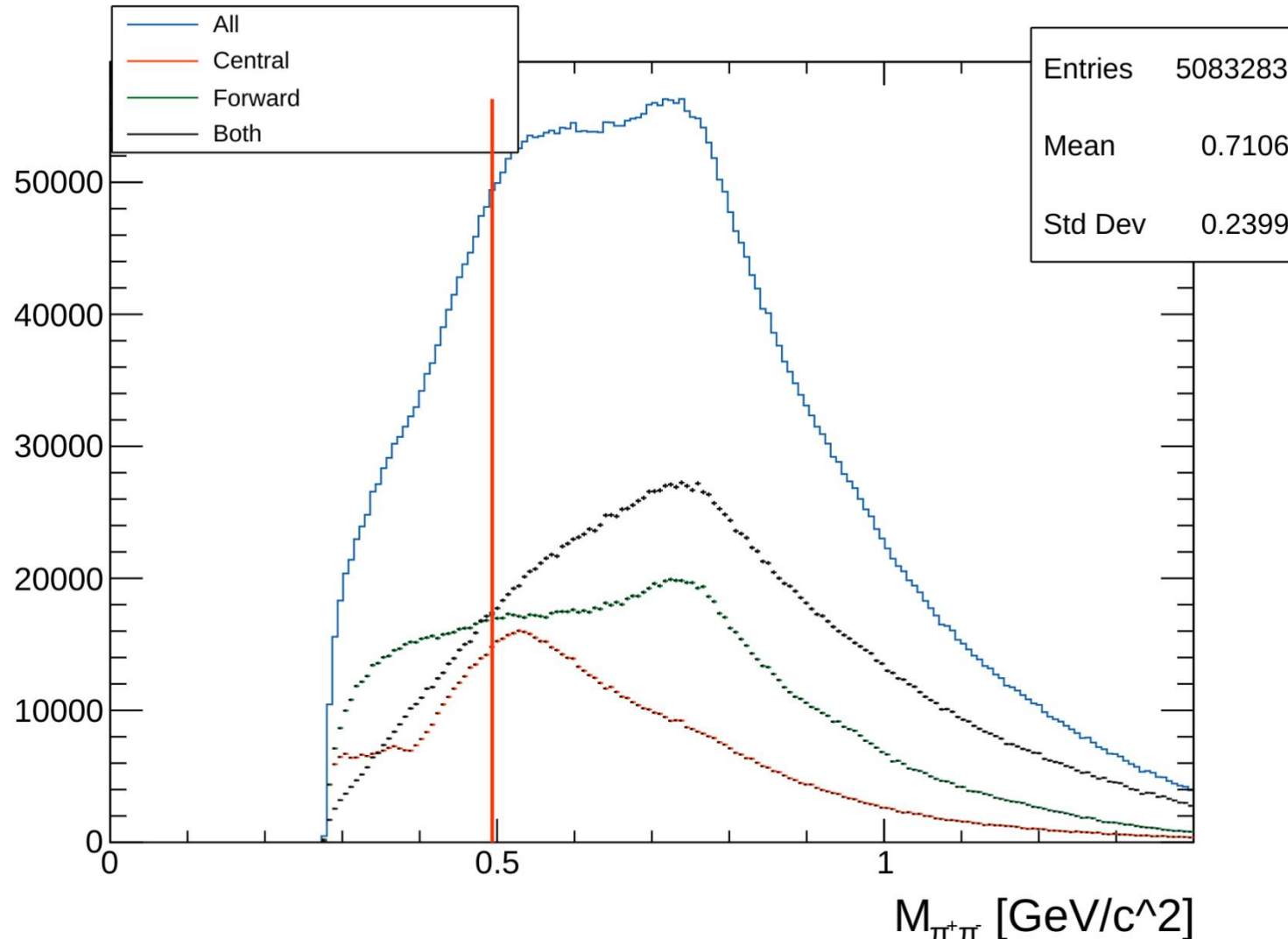
$Q^2 > 1 \text{ GeV}^2$

$W > 2 \text{ GeV}$

$y < 0.85$

RG-D Analysis: K^0

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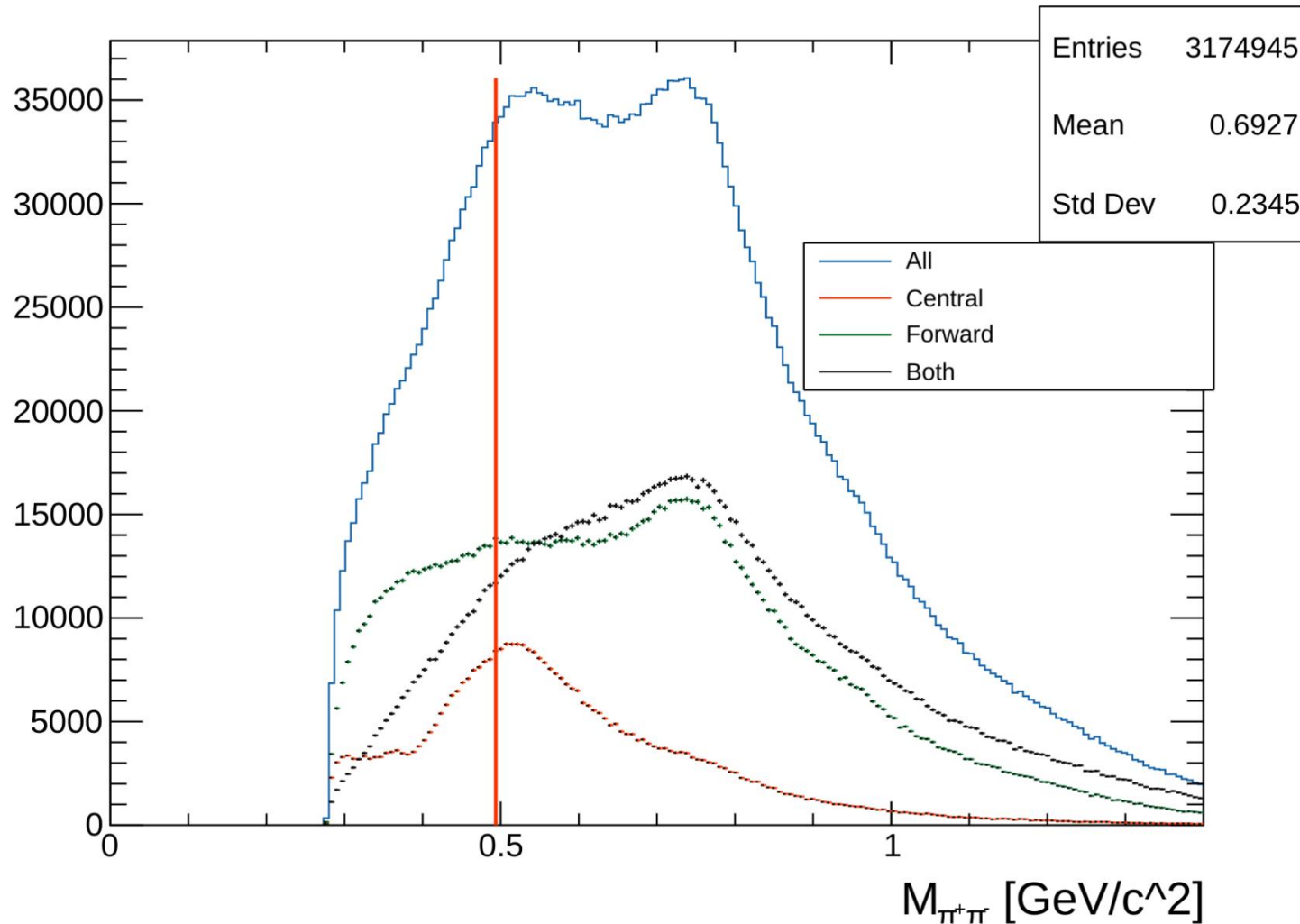
$Q^2 > 1 \text{ GeV}^2$

$W > 2 \text{ GeV}$

$y < 0.85$

RG-D Analysis: K^0

- Find K^0 but only in the central detectors.



Electron :

$-5 < \chi^2_{\text{pid}} < 5$

$-15 < v_z < 5 \text{ cm}$

Pions:

$10 < \chi^2_{\text{pid}} < 10$

Event:

$z_h < 0.7$

$Q^2 > 1 \text{ GeV}^2$

$W > 2 \text{ GeV}$

$y < 0.85$

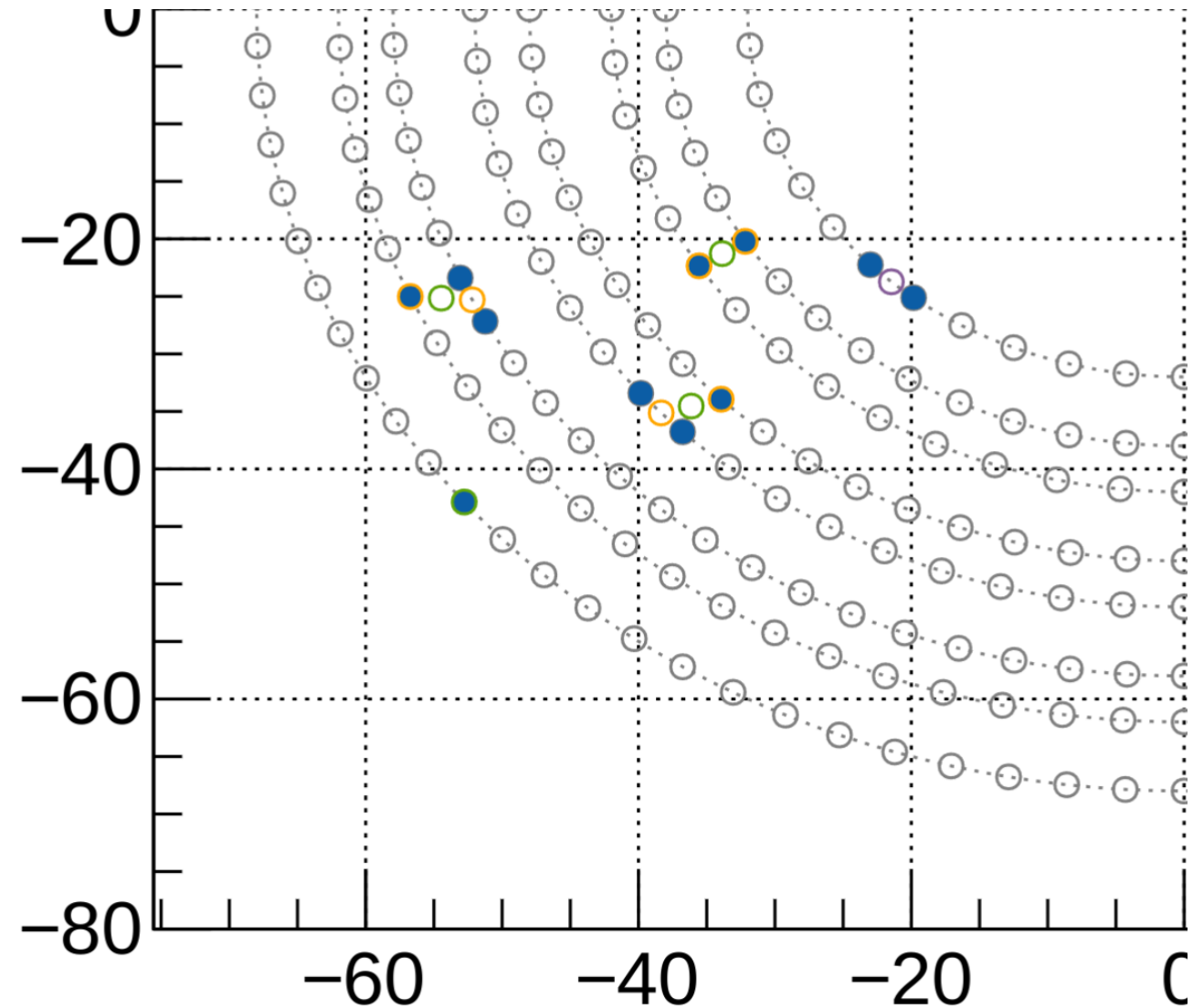
Background merging

- Use trigger-filter script to get background event based on the trigger bit :
 - Use 0x0000008000000000 to select the FC (Faraday Cup) trigger.
 - Give one big file with a specify amout of event. (-n)
 - Need to pick event from the all run range
 - Add separate them depending on the target, magnetic field and current
- Use trigger-split to split the previous output file in N files with 10K event.

AI for ALERT

New clustering:

- Raw hits are the blue dots.
- Merge hits that on the same layer and 1 wire away together into precluster (orange circle).
- Merge precluster that are on the same superlayer and 8mm away into superprecluster (green circle).
- End up with 5 superprecluster that account for track with 7,8 or 9 preclusters.



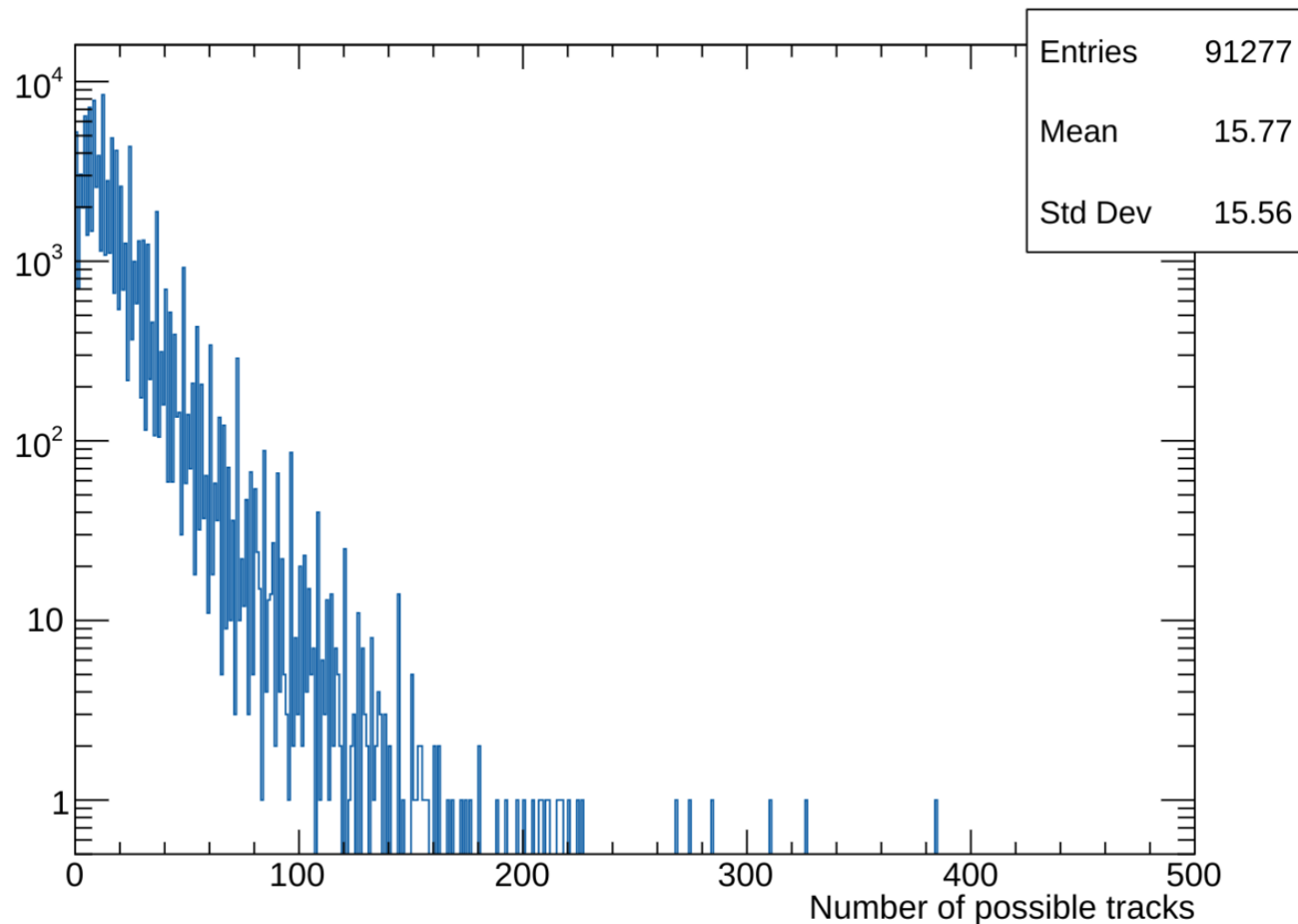
AI for ALERT

Using the new clustering and for 15 random hits, the mean of possible numbers of tracks is 15.

For one event, it can go up to 200 tracks.

For an inference time for the model around $15\mu\text{s}$, we can classify 200 tracks in 3ms.

The time to generate all the possible tracks is, in mean, 0.43 ms.



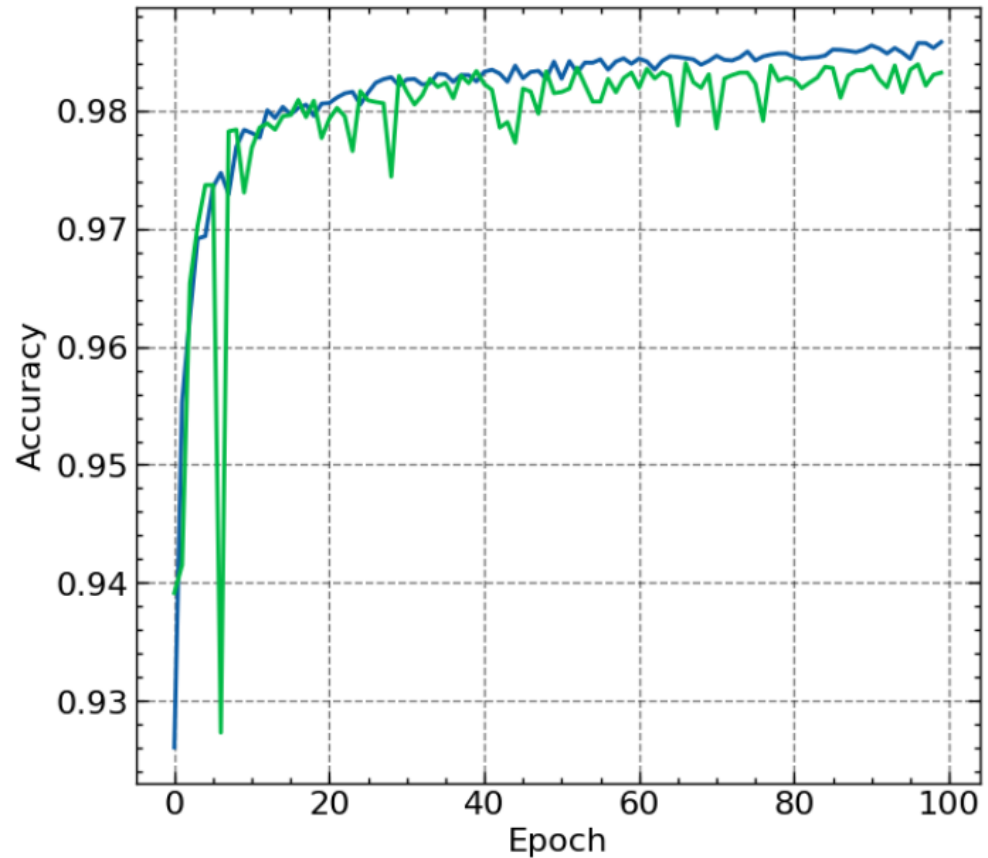
AI for ALERT

Using the new clustering or the preclusters doesn't change the accuracy of the model : 98.1%.

Confusion Matrix

actual values	False	True	
	13540	454	
False	63	14129	
True			
		False	True

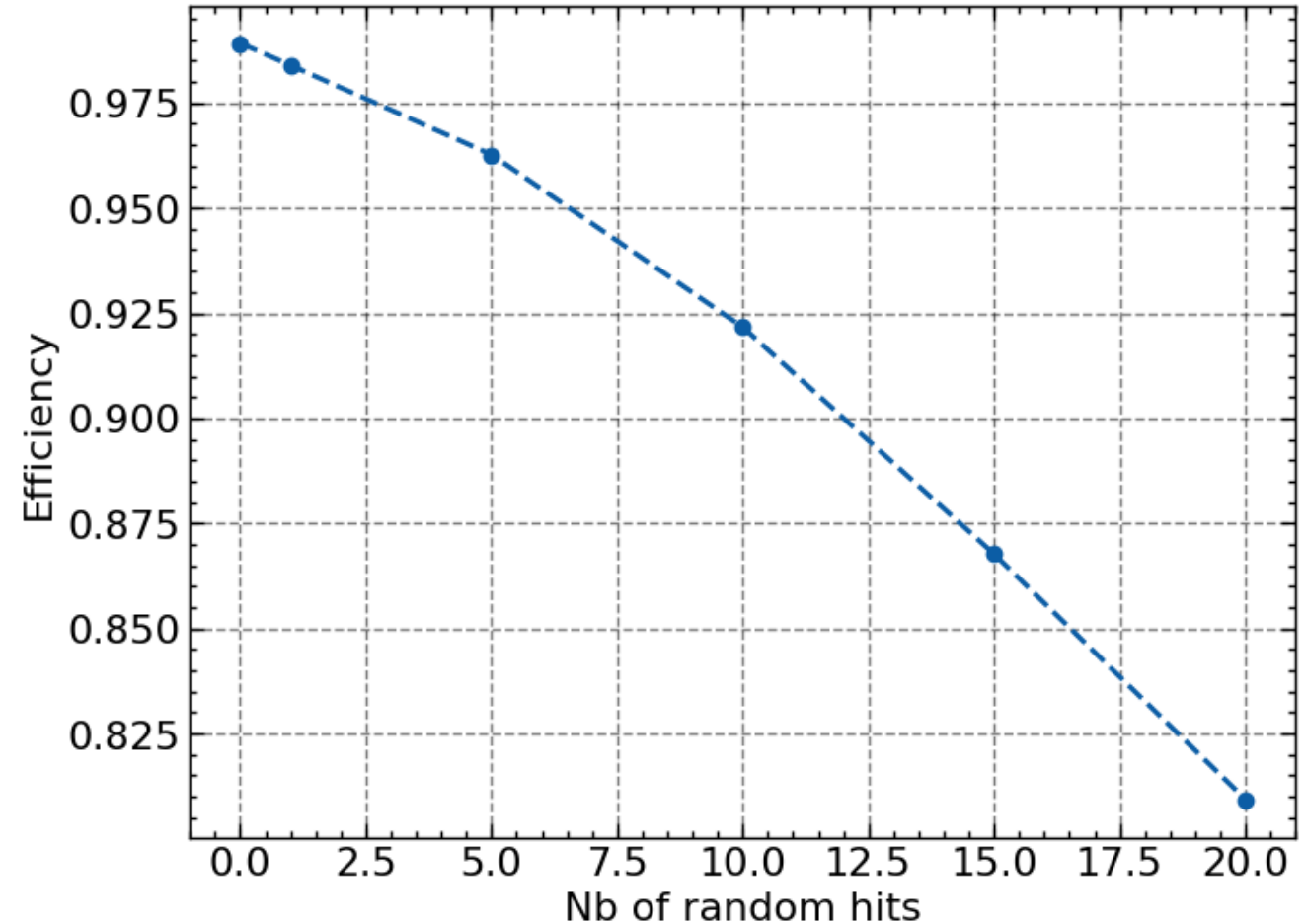
predicted values



AI for ALERT

Efficiency vs nb of random hits added.

Efficiency is nb of time the model find only the good track per event divided by the number of event.



AI for ALERT

- Tasks:
 - Track finding
 - - MLP implementation in java
 - - training and testing with different particles (H, 2H, 3H, 3He, 4He).
 - - testing for different number of tracks (accuracy and efficiency) Udi
 - - MLP vs conventional (efficiency and speed)
 - - Other architecture (almost done, tweak parameters) Udi
 - - GNN change the paradigm no more classifier
 - - PID Udi
 - - Give information from the hits of the hdc (precluster, hit or other ?) + information from the TOF (clustering is not done yet)
 - - Use multiclass classifier to separate H, 2H, 3H, 3He, 4He

To do list:

- RGD:

- Understand why nuclear transparency look like this.
- Sector depend v_x and v_y + fit to cross check, compare to sector independant.
- SIDIS analysis, multiplicity ratio and cut on v_{txy} and v_{tz} and d_{oca} to see if it improve the peak, Δt vs p_{kaon}
- Daniel cross check.

- AI ALERT:

- training and testing with differents particle (H,2H,3H,3He,4He).
- MLP vs conventional (efficiency and speed)

Talk for CLAS collaboration meeting

- RG-D physics
- Run configuration
- Calibration:
 - Beam offset
 - FTOF
 - RF
 - Pass calibration readiness review
- Alignment
- Studies: matt, daniel and me